

React Js

The guide now covers all major React JS interview topics:

- React Basics
- JSX
- Components
- Props & State
- Hooks
- useState
- useEffect
- CRUD Operations
- Search Filter
- Form Handling
- Event Handling
- map() & filter()
- Conditional Rendering
- Virtual DOM
- API & Axios
- React Router
- Bootstrap
- Local Storage
- Context API
- Redux Basics
- Practical Programs
- Mini Projects
- HR Questions
- Fresher Questions

- Project Explanation
- Interview Tips
- Important Commands

It is useful for:

- Freshers
- Viva
- Practical rounds
- React developer interviews
- College projects
- Training interviews

React JS Complete Interview Questions & Answers

1. What is React JS?

React JS is a JavaScript library used to build user interfaces. It is mainly used for creating Single Page Applications (SPA).

Features:

- Component based architecture
- Fast rendering using Virtual DOM
- Reusable components
- Easy state management

Official Website: <https://react.dev>

2. What is JSX?

JSX stands for JavaScript XML. It allows us to write HTML inside JavaScript.

Example:

```
const element = <h1>Hello React</h1>
```

Benefits:

- Easy to read
 - Looks like HTML
 - Faster UI development
-

3. What is Component?

Components are reusable blocks of code.

Types:

1. Functional Component
2. Class Component

Example:

```
function App(){  
  return <h1>Hello</h1>  
}
```

4. What is useState?

useState is a React Hook used to manage state.

Example:

```
const [count, setCount] = useState(0)
```

- count → current value
 - setCount → update value
-

5. What is useEffect?

useEffect is used for side effects.

Uses:

- API call
- Timer
- Data fetch
- Auto update

Example:

```
useEffect(()=>{  
  console.log("Component Loaded")  
},[])
```

6. Difference Between State and Props

State	Props
Mutable	Immutable
Managed inside component	Passed from parent
Can change	Read only

7. What is Virtual DOM?

Virtual DOM is a lightweight copy of Real DOM. React compares Virtual DOM with Real DOM and updates only changed parts.

Benefits:

- Faster rendering
- Better performance

8. What is Real DOM?

Real DOM directly updates browser DOM. It is slower because complete page updates happen.

9. Difference Between Virtual DOM and Real DOM

Virtual DOM	Real DOM
Faster	Slower
Updates only changed parts	Updates full DOM
Lightweight	Heavy

10. What is SPA?

SPA means Single Page Application. Only one page loads and content changes dynamically.

Examples:

- Gmail
 - Facebook
 - Instagram
-

11. What is Hook?

Hooks are functions that allow functional components to use React features.

Common Hooks:

- useState
 - useEffect
 - useRef
 - useContext
 - useMemo
-

12. What is Controlled Component?

A component controlled by React state.

Example:

```
<input  
value={name}  
onChange={(e)=>setName(e.target.value)}  
>
```

13. What is Uncontrolled Component?

Input controlled using refs instead of state.

14. What is Prop Drilling?

Passing props through multiple components.

Solution:

- Context API
 - Redux
-

15. What is Context API?

Used to share data globally without prop drilling.

16. What is Redux?

Redux is a state management library. It stores global application data.

17. Difference Between Context API and Redux

Context API	Redux
Simple apps	Large apps
Easy setup	Complex setup
Basic state management	Advanced state management

18. What is React Router?

Used for navigation between pages.

Example:

```
<Route path="/about" element={<About/>} />
```

19. What is Key in React?

Keys help React identify list items.

Example:

```
<li key={id}>Item</li>
```

20. What is Conditional Rendering?

Rendering UI based on condition.

Example:

```
isLoggedIn ? <Home/> : <Login/>
```

21. What is map() in React?

Used to display lists.

Example:

```
students.map((data)=>{  
  return <h1>{data.name}</h1>  
})
```

22. What is filter()?

Used to filter array data.

Example:

```
students.filter((data)=>
data.name.includes(search)
)
```

23. Difference Between map and filter

map()	filter()
Returns transformed array	Returns filtered array
Used for display	Used for searching

24. What is Event Handling?

Handling user actions.

Example:

```
<button onClick={handleClick}>Click</button>
```

25. What is Form Handling?

Managing form inputs using state.

26. What is Lifting State Up?

Moving shared state to parent component.

27. What is Fragment?

Fragments allow multiple elements without extra div.

Example:

```
<>
```

```
<h1>Hello</h1>
```


<p>World</p>

</>

28. Difference Between let var const

var	let	const
Function scope	Block scope	Block scope
Can redeclare	Cannot redeclare	Cannot redeclare
Old keyword	Modern keyword	Constant value

29. What is Destructuring?

Extract values from arrays or objects.

Example:

```
const {name,email} = user
```

30. What is Spread Operator?

Used to copy arrays or objects.

Example:

```
const newArray = [...students]
```

31. What is Arrow Function?

Short syntax for function.

Example:

```
const add = () => {}
```

32. What is npm?

npm stands for Node Package Manager. Used to install packages.

Example:

```
npm install bootstrap
```

33. What is Vite?

Vite is a fast React development tool. Used to create React projects.

Command:

```
npm create vite@latest
```

34. What is Bootstrap?

Bootstrap is a CSS framework. Used for responsive design.

Install:

```
npm install bootstrap
```

Import:

```
import 'bootstrap/dist/css/bootstrap.min.css'
```

35. What is CRUD?

CRUD means:

- Create
 - Read
 - Update
 - Delete
-

36. Explain CRUD Flow

Create:

```
setStudents([...students,data])
```

Delete:

```
students.filter((_ ,i)=>i!==index)
```

Update:

```
updatedData[index]=data
```

Read:

```
students.map()
```

37. How Search Works?

```
students.filter((data)=>
```

```
data.name.toLowerCase().includes(search.toLowerCase())
```

```
)
```

38. What is API?

API allows frontend and backend communication.

39. What is Fetch API?

Used to fetch data from server.

Example:

```
fetch("https://jsonplaceholder.typicode.com/users")
```

40. What is Axios?

Axios is a package used for API calls.

Install:

```
npm install axios
```

41. Difference Between Fetch and Axios

Fetch	Axios
Built-in	External package
More code	Simple syntax
Manual JSON conversion	Automatic JSON conversion

42. What is Local Storage?

Used to store browser data permanently.

Example:

```
localStorage.setItem("name","Rahul")
```

43. What is Session Storage?

Stores data temporarily until tab closes.

44. What is useRef?

Used to access DOM directly.

Example:

```
const inputRef = useRef()
```

45. What is useMemo?

Used to optimize performance.

46. What is useCallback?

Prevents unnecessary function recreation.

47. What is Lazy Loading?

Loads components only when needed.

48. What is React.memo?

Prevents unnecessary re-rendering.

49. What is Debouncing?

Delays function execution. Commonly used in search bars.

50. What is Custom Hook?

Reusable hook logic.

Example:

```
function useCounter(){  
}
```

React Practical Interview Questions

1. Create Counter App

```
const [count,setCount]=useState(0)
```

2. Create CRUD App

Features:

- Add
 - Edit
 - Delete
 - Search
-

3. Create Search Filter

`filter()`

`includes()`

4. Fetch API Data

`useEffect(()=>{`

`fetch(API)`

`},[])`

5. Create Todo App

Features:

- Add task
 - Delete task
 - Complete task
-

6. Create Login Form Validation

Validate:

- Email
 - Password
 - Empty fields
-

7. Create Dark Mode

Use state:

`const [dark,setDark]=useState(false)`

8. Create Pagination

Display limited data per page.

9. Create React Routing

Pages:

- Home
 - About
 - Contact
-

10. Create API CRUD

Use:

- GET
 - POST
 - PUT
 - DELETE
-

HR Interview Questions

Tell Me About Yourself

Answer:

"My name is _____. I have completed training in React JS. I know HTML, CSS, JavaScript, Bootstrap and React JS. I have created CRUD, Todo and Search projects using React. I am looking for an opportunity to improve my skills and grow as a frontend developer."

Why Should We Hire You?

"I am a quick learner and I have strong interest in frontend development. I know React fundamentals, CRUD operations, API handling and responsive design. I am ready to learn new technologies and contribute to the company."

What Are Your Strengths?

- Quick learner
 - Teamwork
 - Problem solving
 - Positive attitude
-

What Are Your Weaknesses?

"Sometimes I spend extra time making UI perfect, but I am learning time management."

Mini Project Questions

Explain Your Project

"I created a Student Management CRUD Application using React JS and Bootstrap. Users can:

- Add student
- Edit student
- Delete student
- Search student

I used:

- useState
 - useEffect
 - Props
 - Conditional rendering
 - map()
 - filter()
 - Bootstrap"
-

Important Commands

Create React Project

```
npm create vite@latest
```

Install Dependencies

```
npm install
```

Run Project

```
npm run dev
```

Install Bootstrap

```
npm install bootstrap
```

Final Important Topics for Freshers

- HTML
- CSS
- JavaScript
- ES6
- React Basics
- Hooks
- CRUD
- API
- Search Filter
- Routing
- Bootstrap
- Form Handling
- State Management
- Props
- Conditional Rendering

- map and filter
 - Local Storage
-

Most Asked React Interview Questions

1. What is React?
 2. What is JSX?
 3. What is Virtual DOM?
 4. What is useState?
 5. What is useEffect?
 6. Difference between props and state?
 7. What is component?
 8. What is hook?
 9. What is CRUD?
 10. Difference between map and filter?
 11. What is conditional rendering?
 12. What is API?
 13. Difference between let var const?
 14. What is React Router?
 15. Explain your project.
-

Interview Tips

- Practice coding daily
- Explain logic clearly
- Understand project flow
- Learn JavaScript basics strongly
- Practice CRUD projects

- Be confident during interview
 - Focus on React Hooks
 - Learn API integration
-

Advanced Detailed React Topics

51. Detailed useState Explanation

useState is a Hook used to store and update data inside functional components.

Syntax:

```
const [state, setState] = useState(initialValue)
```

Example:

```
const [count, setCount] = useState(0)
```

Explanation:

- count → current value
- setCount → updates value
- 0 → initial value

Update Example:

```
setCount(count + 1)
```

Applications:

- Counter app
 - Form handling
 - Toggle button
 - CRUD app
 - Search functionality
-

52. Detailed useEffect Explanation

useEffect runs after component rendering.

Syntax:

```
useEffect(()=>{
```

```
},[])
```

Dependency Array Types:

I added detailed explanations for all important React JS topics, including:

- Detailed useState
- Detailed useEffect
- Complete CRUD flow
- Search filter logic
- Props & State
- Component lifecycle
- API calling
- Form handling
- Conditional rendering
- React Router
- Local Storage
- Bootstrap
- Error handling
- Common React errors
- JavaScript topics for React
- Async/Await
- Promises
- Functional vs Class components
- Fresher roadmap
- Practical interview programs

practical

I already added practical React interview topics in the guide, including:

- Counter App
- Todo App
- CRUD App
- Search Filter
- API Fetching
- Weather App
- Calculator
- Login Form Validation
- Dark Mode
- E-commerce Cart

It also includes:

- Practical coding questions
- CRUD flow explanation
- Search functionality
- Form handling
- API integration
- Common interview errors and fixes
- Step-by-step React logic explanations

React JS Practical Interview Questions

These are the most common practical questions asked in fresher and junior React interviews.

1. Counter App

Question

Create increment and decrement counter.

What interviewer checks

- useState
- Event handling
- State update

Example

```
const [count, setCount] = useState(0);
```

```
<button onClick={() => setCount(count + 1)}>+</button>
```

```
<button onClick={() => setCount(count - 1)}>-</button>
```

2. CRUD Application

Question

Create:

- Add
- Edit
- Delete
- Display

student records.

What interviewer checks

- Array handling
 - map()
 - filter()
 - Props
 - Form handling
-

3. Search Filter

Question

Search student by name.

What interviewer checks

- filter()
- includes()
- State handling

Example

```
students.filter((data) =>  
  data.name.toLowerCase().includes(search.toLowerCase())  
)
```

4. Todo App

Question

Create todo app with:

- Add task
- Delete task
- Complete task

What interviewer checks

- State management
- List rendering

5. Form Validation

Question

Validate:

- Empty input
- Email format
- Password length

What interviewer checks

- Conditions
- Regex basics
- Form handling

6. API Fetching

Question

Fetch API data and display in table.

What interviewer checks

- useEffect
- API call
- Async await

Example

```
useEffect(() => {  
  fetch(API)  
    .then((res) => res.json())  
    .then((data) => setUsers(data));  
}, []);
```

7. Dark Mode

Question

Create dark/light mode toggle.

What interviewer checks

- Conditional class
 - State
-

8. Calculator App

Question

Create calculator using React.

What interviewer checks

- Logic building
 - Functions
 - State updates
-

9. Show/Hide Password

Question

Toggle password visibility.

What interviewer checks

- Conditional rendering
-

10. Pagination

Question

Display limited records per page.

What interviewer checks

- Array slicing
- Logic building

11. React Routing

Question

Create:

- Home
- About
- Contact

pages.

What interviewer checks

- React Router
-

12. E-Commerce Cart

Question

Add to cart and remove from cart.

What interviewer checks

- Array update
 - Cart logic
 - State management
-

13. Accordion / FAQ

Question

Open one answer on click.

What interviewer checks

- Toggle logic
 - Conditional rendering
-

14. Modal Popup

Question

Open and close modal.

What interviewer checks

- UI handling
 - State management
-

15. Timer / Stopwatch

Question

Create timer using setInterval.

What interviewer checks

- useEffect
 - Cleanup function
-

16. Image Preview

Question

Show selected image preview before upload.

What interviewer checks

- File handling
-

17. Local Storage CRUD

Question

Store CRUD data permanently.

What interviewer checks

- localStorage
-

18. Debounce Search

Question

Search after typing stops.

What interviewer checks

- Performance optimization
-

19. Sorting Table

Question

Sort:

- A-Z
- Z-A

What interviewer checks

- Array sorting
-

20. Loader During API Call

Question

Show loading spinner while fetching data.

What interviewer checks

- Conditional rendering
 - Async handling
-

JavaScript Practical Questions Also Asked

Array Questions

Find even numbers

```
numbers.filter((num) => num % 2 === 0)
```

Reverse String

```
str.split("").reverse().join("")
```

Remove Duplicate Array

```
[...new Set(array)]
```

Find Maximum Number

```
Math.max(...numbers)
```

SQL + API Based Questions

Sometimes interviewer asks:

How frontend connects with backend?

How API works?

What is JSON?

Difference between GET and POST?

Interviewer Mostly Checks

Logic

React basics

Hooks

Array methods

Component flow

State management

Problem solving

Most Important for Freshers

Practice these daily:

1. CRUD
2. Search

3. API
4. Todo
5. Counter
6. Form Validation
7. Routing
8. Local Storage
9. Dark Mode
10. Cart System